

# Cell-State Reachability

MEASURED → MODELLED → VALIDATE

Point-estimate cone geometry — not a calibrated state-conversion claim

1

Inputs

source objects reused from Zhu workflow

TARGET DIRECTION **d**

Source-study-reused population contrast (not independent or a trajectory)

EFFECT DICTIONARY **E**

Screen-derived CRISPRi DE z-scores in post-expansion Rest CD4 cells

Target union: 25,672; shared screen: 8,950

Final merged analysis: 6,188 genes

Dictionary is donor-collapsed; not measured in polarized Th2 cells

2

Projection

what the model computes

The diagram illustrates the projection of a target vector onto a cone of model fits. A black vector labeled 'target' originates from a common point. A light blue cone, labeled 'projected model fit', also originates from this point. A green vector within the cone represents the 'projected model fit'. A brown vector, labeled 'unmatched residual', connects the tip of the target vector to the tip of the projected model fit vector.

Non-negative linear combinations in declared effect space

Residual is model-relative

3

Challenge

source-bound + measured transfer tests

0.444 ± 0.018

mean ± SD, 12 hash-frozen gene splits

selected population contrast; not trajectory

A dot plot with a horizontal axis. There are 12 vertical tick marks. A thick vertical line is positioned at the center, representing the mean. The range of the data is indicated by the spread of the tick marks.

range: 0.417-0.473

Frozen transfer challenges

Source gain +0.087 / +0.090

Donor-pair gain +0.032; nRMSE 1.153 vs 1.018

Released positional: cone 0.251 within → -0.019 reciprocal

Δ -0.291; 3/12 beat best single

Arce ranks 0.148/0.084/0.088 | Zhu 9/9; cosine 0.580

Schmidt p(top-200): donor 0.89 | donor+M/L 0.30 | donor+ctx 0.75

RNA→flow 0.72-0.85; descriptive, not donor/guide/state validation

4

Limits

what remains unresolved

REPLICATION GATE

Donor-resolved, polarized Th2 starting-state study

CALIBRATION GATE

Structured holdouts, uncertainty + matched baselines

NOT ESTABLISHED

State conversion, rescue, or target validation

Goudy cross-experiment stress: cone 0.095;

triple donor cosine 0.048 — model fails

## CLAIM BOUNDARY

Point-estimate directional geometry • correlated holdout diagnostic • model cone ≠ biological reachability